

Задача 6.

9.1.27

$$\int_1^{e^2} \frac{\ln^3 x}{3x} dx = \frac{1}{3} \int_1^{e^2} \frac{\ln^3 x}{x} dx = \left[\begin{array}{l} t = \ln x \quad dt = \frac{1}{x} dx \\ dx = x dt \end{array} \right.$$

$$\left. \begin{array}{l} x = e^2 \quad t = 2 \\ x = 1 \quad t = 0 \end{array} \right] = \frac{1}{3} \int_0^2 t^3 dt = \frac{1}{3} \cdot \frac{t^4}{4} \Big|_0^2 =$$

$$= \frac{1}{12} \cdot t^4 \Big|_0^2 = \frac{1}{12} (2^4 - 0) = \frac{1}{12} (16 - 0) =$$

$$= \frac{1}{12} \cdot 16 = \frac{14}{12} = \frac{7}{6}$$

9.1.28

$$\int_{\pi}^{2\pi} \frac{x + \cos x}{x^2 + 2 \sin x} dx = \left[\begin{array}{l} t = x^2 + 2 \sin x \quad dt = (2x + 2 \cos x) dx = \\ = 2(x + \cos x) dx \quad dx = \frac{1}{2(x + \cos x)} dt \end{array} \right.$$

$$\left. \begin{array}{l} x = 2\pi \quad t = 4\pi^2 \\ x = \pi \quad t = \pi^2 \end{array} \right] = \int_{\pi^2}^{4\pi^2} \frac{dt}{2t} = \frac{1}{2} \ln |t| \Big|_{\pi^2}^{4\pi^2} =$$

$$= \frac{1}{2} (\ln 4\pi^2 - \ln \pi^2) = \frac{1}{2} \ln \frac{4\pi^2}{\pi^2} = \frac{1}{2} \ln 4 =$$

$$= \frac{1}{2} \cdot 2 \ln 2 = \ln 2$$

9.1.29

$$\int_0^1 \frac{4 \operatorname{arctg} x - x}{1+x^2} dx = 4 \int_0^1 \frac{\operatorname{arctg} x}{1+x^2} dx - \int_0^1 \frac{x}{1+x^2} dx =$$

$$= \left[\begin{array}{l} t = \operatorname{arctg} x \quad dt = \frac{1}{1+x^2} dx \\ dx = 1+x^2 dt \end{array} \right. \quad \left. \begin{array}{l} x=1 \quad t = \frac{\pi}{4} \\ x=0 \quad t = 0 \end{array} \right] = \left[\begin{array}{l} u = 1+x^2 \quad du = 2x dx \\ dx = \frac{1}{2x} du \end{array} \right.$$

$$\begin{aligned}
 \left. \begin{array}{l} x=1 \quad u=2 \\ x=0 \quad u=1 \end{array} \right] &= 4 \cdot \int_0^{\frac{\pi}{4}} t \, dt - \int_1^2 \frac{1}{u} \, du = \\
 &= 4 \cdot \frac{t^2}{2} \Big|_0^{\frac{\pi}{4}} - \frac{1}{2} \int_1^2 \frac{1}{u} \, du = 2 \cdot t^2 \Big|_0^{\frac{\pi}{4}} - \\
 &= \frac{1}{2} \ln|u| \Big|_1^2 = 2 \left(\left(\frac{\pi}{4} \right)^2 - 0^2 \right) - \frac{1}{2} (\ln 2 - \ln 1) = \\
 &= 2 \left(\frac{\pi^2}{16} - 0 \right) - \frac{1}{2} (\ln 2 - 0) = \frac{\pi^2}{8} - \frac{\ln 2}{2}
 \end{aligned}$$

9.1.30

$$\begin{aligned}
 \int_1^e \frac{\sin \ln x}{x} \, dx &= \left[\begin{array}{l} t = \ln x \quad dt = \frac{1}{x} dx \\ dx = x \, dt \end{array} \quad \begin{array}{l} x=e \quad t=1 \\ x=1 \quad t=0 \end{array} \right] = \\
 &= \int_0^1 \sin t \, dt = \cos t \Big|_0^1 = (\cos 1 - \cos 0) = \cos 1 - 1
 \end{aligned}$$

9.1.31.

$$\begin{aligned}
 \int_{-1}^0 \frac{3^x - 2^x}{6^x} \, dx &= \int_{-1}^0 \frac{3^x}{6^x} \, dx - \int_{-1}^0 \frac{2^x}{6^x} \, dx = \int_{-1}^0 \frac{1}{2^x} \, dx - \int_{-1}^0 \frac{1}{3^x} \, dx = \\
 &= \int_{-1}^0 \left(\frac{1}{2} \right)^x \, dx - \int_{-1}^0 \left(\frac{1}{3} \right)^x \, dx = \frac{\left(\frac{1}{2} \right)^x}{\ln \frac{1}{2}} \Big|_{-1}^0 - \frac{\left(\frac{1}{3} \right)^x}{\ln \frac{1}{3}} \Big|_{-1}^0 = \\
 &= -\frac{1}{2^x \ln 2} \Big|_{-1}^0 + \frac{1}{3^x \ln 3} \Big|_{-1}^0 = -\left(\frac{1}{2^0 \ln 2} - \frac{1}{2^{-1} \ln 2} \right) + \left(\frac{1}{3^0 \ln 3} - \frac{1}{3^{-1} \ln 3} \right) = \\
 &= -\left(\frac{1}{\ln 2} - \frac{2}{\ln 2} \right) + \left(\frac{1}{\ln 3} - \frac{3}{\ln 3} \right) = \frac{2}{\ln 2} - \frac{1}{\ln 2} + \frac{1}{\ln 3} - \frac{3}{\ln 3} = \\
 &= \frac{1}{\ln 2} - \frac{2}{\ln 3}
 \end{aligned}$$

g. 1.32

$$\int_0^1 \frac{x dx}{\sqrt{x^4 + x^2 + 1}} = \left[\begin{array}{l} t = x^2 \quad dt = 2x dx \\ dx = \frac{1}{2x} dt \end{array} \quad \begin{array}{l} x=1 \quad t=1 \\ x=0 \quad t=0 \end{array} \right] =$$

$$= \int_0^1 \frac{dt}{2\sqrt{t^2 + t + 1}} = \frac{1}{2} \int_0^1 \frac{dt}{\sqrt{(t + \frac{1}{2})^2 + \frac{3}{4}}} =$$

$$= \left[\begin{array}{l} u = t + \frac{1}{2} \quad du = dt \\ x=1 \quad t=1.5 \quad x=0 \quad t=0.5 \end{array} \right] = \frac{1}{2} \int_{0.5}^{1.5} \frac{du}{\sqrt{u^2 + \frac{3}{4}}} =$$

$$= \frac{1}{2} \int_{0.5}^{1.5} \frac{du}{\sqrt{u^2 + (\frac{\sqrt{3}}{2})^2}} = \frac{1}{2} \cdot \frac{2}{\sqrt{3}} \operatorname{arctg} \frac{2u}{\sqrt{3}} \Big|_{0.5}^{1.5} =$$

$$= \frac{\sqrt{3}}{3} \left(\operatorname{arctg} \frac{2 \cdot 1.5}{\sqrt{3}} - \operatorname{arctg} \frac{2 \cdot 0.5}{\sqrt{3}} \right) = \frac{\sqrt{3}}{3} \left(\operatorname{arctg} \frac{3\sqrt{3}}{3} - \operatorname{arctg} \frac{\sqrt{3}}{3} \right) =$$

$$= \frac{\sqrt{3}}{3} \left(\frac{\pi}{3} - \frac{\pi}{6} \right) = \frac{\sqrt{3}}{3} \cdot \frac{\pi}{6} = \frac{\pi\sqrt{3}}{18}$$

g. 1.33

$$\int_0^{\frac{\pi}{4}} \operatorname{tg}^3 x dx = \int_0^{\frac{\pi}{4}} \frac{\sin^3 x}{\cos^3 x} dx = \int_0^{\frac{\pi}{4}} \frac{(1 - \cos^2 x) \sin x}{\cos^3 x} dx =$$

$$= \left[\begin{array}{l} t = \cos x \quad dt = -\sin x dx \\ dx = -\frac{1}{\sin x} dt \end{array} \quad \begin{array}{l} x = \frac{\pi}{4} \quad t = \frac{\sqrt{2}}{2} \\ x = 0 \quad t = 1 \end{array} \right] =$$

$$= - \int_{\frac{\sqrt{2}}{2}}^1 \frac{1-t^2}{t^3} dt = - \left(\int_{\frac{\sqrt{2}}{2}}^1 \frac{1}{t^3} dt - \int_{\frac{\sqrt{2}}{2}}^1 \frac{t^2}{t^3} dt \right) =$$

$$= - \left(\frac{t^{-2}}{-2} \Big|_{\frac{\sqrt{2}}{2}}^1 - \ln|t| \Big|_{\frac{\sqrt{2}}{2}}^1 \right) = \frac{1}{2t^2} \Big|_{\frac{\sqrt{2}}{2}}^1 + \ln|t| \Big|_{\frac{\sqrt{2}}{2}}^1 =$$

$$= \left(\frac{1}{2 \left(\frac{\sqrt{2}}{2} \right)^2} - \frac{1}{2 \cdot 1^2} \right) + \left(\ln \frac{\sqrt{2}}{2} - \ln 1 \right) = \left(1 - \frac{1}{2} \right) + \left(-\frac{1}{2} \ln 2 \right) =$$

$$= \frac{1}{2} - \frac{1}{2} \ln 2 = \frac{1}{2} (1 - \ln 2)$$

9.1.34

$$\int_0^{\frac{\pi}{3}} \lg x \cdot \ln \cos x \, dx = \left[\begin{array}{l} t = \ln \cos x \quad dt = \frac{\sin x}{\cos x} dx \quad x = \frac{\pi}{3} \quad t = -\ln 2 \\ dx = \cos x / \sin x \, dt \quad x = 0 \quad t = 0 \end{array} \right] =$$

$$= \int_0^{-\ln 2} \frac{\sin x}{\cos x} \cdot t \cdot \frac{\cos x}{\sin x} dt = \int_0^{-\ln 2} t \, dt = \frac{t^2}{2} \Big|_0^{-\ln 2}$$

$$= \frac{1}{2} (-\ln^2 2 - 0^2) = -\frac{1}{2} \ln^2 2$$

9.1.35

$$\int_0^2 \frac{x \, dx}{\sqrt{x+2} + \sqrt{3x+2}} = \int_0^2 \frac{x \, dx}{\sqrt{x+2} + \sqrt{3x+2}} \cdot \frac{\sqrt{3x+2} - \sqrt{x+2}}{\sqrt{3x+2} - \sqrt{x+2}} =$$

$$= \int_0^2 \frac{x \sqrt{3x+2} - x \sqrt{x+2}}{3x+2 - x-2} dx = \int_0^2 \frac{x (\sqrt{3x+2} - \sqrt{x+2})}{2x} dx =$$

$$= \frac{1}{2} \left(\int_0^2 \sqrt{3x+2} \, dx - \int_0^2 \sqrt{x+2} \, dx \right) = \left[\begin{array}{l} t = 3x+2 \quad dt = 3 \, dx \quad x=2 \quad t=8 \\ dx = \frac{1}{3} \, dt \quad x=0 \quad t=2 \end{array} \right] =$$

$$= \frac{1}{2} \left(\int_2^8 \frac{\sqrt{t}}{3} \, dt - \int_0^2 \sqrt{x+2} \, dx \right) = \left[\begin{array}{l} u = x+2 \quad du = dx \quad x=2 \quad t=4 \\ x=0 \quad t=2 \end{array} \right] =$$

$$= \frac{1}{2} \left(\int_2^8 \frac{\sqrt{t}}{3} \, dt - \int_2^4 \sqrt{u} \, du \right) =$$

$$= \frac{1}{2} \left(\frac{1}{3} \cdot \frac{2}{3} t^{3/2} \Big|_2^8 - \frac{2}{3} u^{3/2} \Big|_2^4 \right) = \frac{1}{2} \left(\frac{2}{9} (8^{3/2} - 2^{3/2}) - \frac{2}{3} (4^{3/2} - 2^{3/2}) \right) =$$

$$= \frac{1}{2} \left(\frac{2}{9} (16\sqrt{2} - 2\sqrt{2}) - \frac{2}{3} (8 - 2\sqrt{2}) \right) =$$

$$= \frac{1}{2} \left(\frac{2}{9} 14\sqrt{2} - \frac{2}{3} (8 - 2\sqrt{2}) \right) = \frac{14\sqrt{2}}{9} - \frac{16 - 4\sqrt{2}}{3} = \frac{26\sqrt{2} - 48}{9}$$

9.1.66

$$\int_0^{\frac{\pi}{2}} \frac{5 dx}{1 + \cos x} = 5 \int_0^{\frac{\pi}{2}} \frac{1}{2 \cos^2 \frac{x}{2}} dx = 5 \int_0^{\frac{\pi}{2}} \frac{1}{\cos^2 \frac{x}{2}} d\frac{x}{2} =$$

$$= 5 \operatorname{tg} \frac{x}{2} \Big|_0^{\frac{\pi}{2}} = 5 (\operatorname{tg} \frac{\pi}{4} - \operatorname{tg} 0) = 5 (1 - 0) = 5$$

9.1.67.

$$\int_0^{\ln 4} \sqrt{e^x - 1} dx = \int_1^4 \sqrt{t-1} dt$$

$$\left[\begin{array}{l} t = e^x \quad dt = e^x dx \quad x = \ln t \quad t = 4 \\ dx = \frac{1}{e^x} dt \quad x = 0 \quad t = 1 \end{array} \right] = \int_1^4 \frac{\sqrt{t-1}}{t} dt =$$

$$= \left[\begin{array}{l} u = t-1 \\ dt = 2 du \end{array} \quad \begin{array}{l} x = 4 \quad t = 3 \\ x = 1 \quad t = 0 \end{array} \right] = 2 \int_0^3 1 - \frac{1}{u^2 + 1} du =$$

$$= 2 \left(\int_0^3 1 du - \int_0^3 \frac{1}{u^2 + 1} du \right) = 2 \left(u \Big|_0^3 - \frac{1}{1} \operatorname{arctg} \frac{u}{1} \Big|_0^3 \right) =$$

$$= 2 (13 - 0) - (\operatorname{arctg} 3 - \operatorname{arctg} 0) = 6 - 2 \operatorname{arctg} 3 + 2 \operatorname{arctg} 0$$

9.1.68

$$\int_0^{\frac{\pi}{2}} \frac{dx}{1 + \cos x + \sin x} = \left[\begin{array}{l} t = \operatorname{tg} \frac{x}{2} \\ dt = \frac{1}{2 \cos^2 \frac{x}{2}} dx \end{array} \quad \begin{array}{l} \sin x = \frac{2t}{t^2 + 1} \quad x = \frac{\pi}{2} \quad t = 1 \\ \cos x = \frac{1-t^2}{t^2 + 1} \quad x = 0 \quad t = 0 \end{array} \right] =$$

$$= \int_0^1 \frac{dt}{t+1} = \int_0^1 \frac{d(t+1)}{t+1} = \ln |t+1| \Big|_0^1 =$$

$$= \ln 2 - \ln 1 = \ln 2.$$

9.1.69

$$\int_0^{\frac{\pi}{2}} \frac{\cos x \, dx}{6 - 5 \sin x + \sin^2 x} = \left[\begin{array}{l} t = \sin x \\ dt = \cos x \, dx \\ dx = \frac{1}{\cos x} dt \\ x = 0 \quad t = 0 \\ x = \frac{\pi}{2} \quad t = 1 \end{array} \right] =$$

$$= \int_0^1 \frac{dt}{6 - 5t + t^2} = \int_0^1 \frac{dt}{(t-3)(t-2)} =$$

$$\left[\frac{A}{t-3} + \frac{B}{t-2} \right] \quad \frac{1}{(t-3)(t-2)} = \frac{A}{t-3} + \frac{B}{t-2}$$

$$1 = A(t-2) + B(t-3)$$

$$1 = At - 2A + Bt - 3B$$

$$1 = At + Bt - 2A - 3B$$

$$1 = t(A+B) + (-2A-3B)$$

$$\begin{cases} 1 = -2A - 3B \\ 0 = A + B \end{cases} \quad \begin{cases} -2(-B) - 3B = 1 \\ A = -B \end{cases} \quad \begin{cases} -B = 1 \\ A = -B \end{cases} \quad \begin{cases} B = -1 \\ A = 1 \end{cases}$$

$$\left[\frac{1}{t-3} - \frac{1}{t-2} \right] =$$

$$= \int_0^1 \frac{1}{t-3} - \frac{1}{t-2} dt = \int_0^1 \frac{dt}{t-3} - \int_0^1 \frac{dt}{t-2} =$$

$$= \int_0^1 \frac{d(t-3)}{t-3} - \int_0^1 \frac{d(t-2)}{t-2} = \ln|t-3| \Big|_0^1 - \ln|t-2| \Big|_0^1 =$$

$$= (\ln 2 - \ln 3) - (\ln 1 - \ln 2) =$$

$$= \ln 2 - \ln 3 - \ln 1 + \ln 2 = 2\ln 2 - \ln 3$$

9.1.70

$$\int_1^2 3x(1-x)^{17} dx = 3 \int_1^2 x(1-x)^{17} dx \quad \begin{matrix} t=1-x \\ dt=-dx \end{matrix}$$

$$\int_1^2 3x(1-x)^{17} dx =$$

$$\int_1^2 3x(1-x)^{17} dx = 3 \int_1^2 x(1-x)^{17} dx = \left[\begin{matrix} t=1-x & x=1-t \\ dx=-dt \end{matrix} \right] \begin{matrix} x=2 \\ t=-1 \\ x=1 \\ t=0 \end{matrix}$$

$$= 3 \int_{-1}^0 -(1-t)t^{17} dt = -3 \int_{-1}^0 t^{17} - t^{18} dt =$$

$$= -3 \left(\int_{-1}^0 t^{17} dt - \int_{-1}^0 t^{18} dt \right) = -3 \left(\left. \frac{t^{18}}{18} \right|_{-1}^0 - \left. \frac{t^{19}}{19} \right|_{-1}^0 \right) =$$

$$= -3 \left(\left(\frac{0^{18}}{18} - \frac{(-1)^{18}}{18} \right) - \left(\frac{0^{19}}{19} - \frac{(-1)^{19}}{19} \right) \right) =$$

$$= -3 \left(\left(0 - \frac{1}{18} \right) - \left(0 + \frac{1}{19} \right) \right) = -3 \left(-\frac{1}{18} - \frac{1}{19} \right) =$$

$$= \frac{3}{18} + \frac{3}{19} = \frac{1}{6} + \frac{3}{19} = \frac{37}{114}$$

9.1.100

$$\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \frac{x dx}{\sin^2 x} = \left[\begin{matrix} u=x & du=dx \\ dv=\frac{dx}{\sin^2 x} & v=-\operatorname{ctg} x \end{matrix} \right] =$$

$$= -x \operatorname{ctg} x \Big|_{\frac{\pi}{6}}^{\frac{\pi}{2}} - \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} -\operatorname{ctg} x dx = \left(-\frac{\pi}{2} \operatorname{ctg} \frac{\pi}{2} + \frac{\pi}{6} \operatorname{ctg} \frac{\pi}{6} \right) +$$

$$+ \ln |\sin x| \Big|_{\frac{\pi}{6}}^{\frac{\pi}{2}} = \left(0 + \frac{\pi}{6} \sqrt{3} \right) + \left(\ln |\sin \frac{\pi}{2}| - \ln |\sin \frac{\pi}{6}| \right) =$$

$$= \frac{\pi \sqrt{3}}{6} + \left(0 - \ln \frac{1}{2} \right) = \frac{\pi \sqrt{3}}{6} + \ln 2$$

9.1.101.

$$\int_0^{0,2} x e^{5x} dx = \left[\begin{array}{l} u = x \quad du = dx \\ dv = e^{5x} dx \quad v = e^{5x} \frac{1}{5} \end{array} \right] =$$

$$= x \cdot \frac{1}{5} e^{5x} \Big|_0^{0,2} - \int_0^{0,2} e^{5x} \cdot \frac{1}{5} dx =$$

$$= \frac{1}{5} (0,2 \cdot e^{5 \cdot 0,2} - 0) - \frac{1}{5} \cdot \frac{1}{5} \cdot e^{5x} \Big|_0^{0,2} =$$

$$= \frac{1}{5} \cdot 0,2 \cdot e - \frac{1}{25} (e^{5 \cdot 0,2} - e^{5 \cdot 0}) =$$

$$= \frac{e}{25} - \frac{1}{25} (e - 1) = \frac{e}{25} - \frac{e}{25} + \frac{1}{25} = \frac{1}{25} = 0,04.$$

9.1.102. —

$$\int_{\pi/3}^{\pi/2} 4x \operatorname{tg}^2 x dx = \int_{\pi/3}^{\pi/2} x \cdot \operatorname{tg}^2 x dx = \left[\begin{array}{l} u = x \quad du = dx \\ dv = \operatorname{tg}^2 x dx \quad v = -x + \frac{\sin x}{\cos x} \end{array} \right] =$$

$$= x \cdot \left(\frac{\sin x}{\cos x} - x \right) \Big|_{\pi/3}^{\pi/2} - \int_{\pi/3}^{\pi/2} -x + \frac{\sin x}{\cos x} dx =$$

$$= 1) \quad x \cdot \left(\frac{\sin x}{\cos x} - x \right) \Big|_{\pi/3}^{\pi/2} = \frac{1}{2} x \operatorname{tg} x - 2x \Big|_{\pi/3}^{\pi/2} =$$

$$= \left(\frac{\pi}{2} \operatorname{tg} \frac{\pi}{2} - 2 \frac{\pi}{2} \right) - \left(\frac{\pi}{3} \operatorname{tg} \frac{\pi}{3} - 2 \frac{\pi}{3} \right) =$$

9.1.103

$$\int_1^{e^2} \ln^2 x \, dx = \left[\begin{array}{ll} u = \ln^2(x) & du = 2 \frac{\ln x}{x} dx \\ dv = dx & v = x \end{array} \right] =$$

$$= x \ln^2 x \Big|_1^{e^2} - \int_1^{e^2} x \cdot 2 \frac{\ln x}{x} dx = x \ln^2 x \Big|_1^{e^2} - 2 \int_1^{e^2} \ln x \, dx =$$

$$= 1) \quad x \ln^2 x \Big|_1^{e^2} = e^2 \ln^2 e^2 - \ln^2 1 = 2e^2 - 0 = 4e^2$$

$$2) \quad \int_1^{e^2} \ln x \, dx = \left[\begin{array}{ll} u = \ln x & du = \frac{1}{x} dx \\ dv = 2dx & v = 2x \end{array} \right] =$$

$$= \ln x \cdot 2x \Big|_1^{e^2} - \int_1^{e^2} 2x \cdot \frac{1}{x} dx = 2x \ln x \Big|_1^{e^2} - 2 \int_1^{e^2} 1 \, dx =$$

$$= 2x \ln x \Big|_1^{e^2} - 2x \Big|_1^{e^2} = (2e^2 \ln e^2 - 2 \ln 1) -$$

$$- (2e^2 - 2) = (4e^2 - 0) - (2e^2 - 2) = 4e^2 - 2e^2 + 2 =$$

$$= 2e^2 + 2$$

$$= 4e^2 - 2e^2 + 2 = 2e^2 + 2 = 2(e^2 + 1)$$

9.106

$$\int_0^{\sqrt{3}} \frac{x^2}{(1+x^2)^2} dx = \left[\begin{array}{ll} u = x & du = dx \\ dv = \frac{x}{(1+x^2)^2} dx & v = -\frac{1}{2} \frac{1}{x^2+1} \end{array} \right] =$$

$$= x \cdot \left(-\frac{1}{2} \cdot \frac{1}{x^2+1} \right) \Big|_0^{\sqrt{3}} - \int_0^{\sqrt{3}} -\frac{1}{2} \cdot \frac{1}{x^2+1} dx =$$

$$= -\frac{1}{2} \frac{x}{x^2+1} \Big|_0^{\sqrt{3}} + \frac{1}{2} \cdot \frac{1}{1} \operatorname{arctg} x \Big|_0^{\sqrt{3}} =$$

$$= -\frac{1}{2} \left(\frac{\sqrt{3}}{3+1} - 0 \right) + \frac{1}{2} (\operatorname{arctg} \sqrt{3} - \operatorname{arctg} 0) =$$

$$= -\frac{1}{2} \cdot \frac{\sqrt{3}}{4} + \frac{1}{2} \cdot \frac{\pi}{3} = \frac{\pi}{6} - \frac{\sqrt{3}}{8}$$

9.1.105

$$\int_0^2 \frac{x^3}{\sqrt{1+x^2}} dx = \left[\begin{array}{l} u = x^2 \quad du = 2x dx \\ dv = \frac{x}{\sqrt{1+x^2}} dx \quad v = \int \frac{1}{\sqrt{1+x^2}} dx = \\ \left[\begin{array}{l} t = 1+x^2 \\ dt = 2x dx \end{array} \right] = \int \frac{dt}{2\sqrt{t}} = \frac{1}{2} 2\sqrt{t} + C = \sqrt{1+x^2} + C \end{array} \right] =$$

$$= x^2 \sqrt{1+x^2} \Big|_0^2 - \int_0^2 \sqrt{1+x^2} \cdot 2x dx =$$

$$= \left[\int_0^2 \sqrt{1+x^2} \cdot 2x dx = \left[\begin{array}{l} t = 1+x^2 \quad dt = 2x dx \\ dx = \frac{1}{2x} dx \end{array} \right] \begin{array}{l} x=2 \quad t=5 \\ x=0 \quad t=1 \end{array} \right] =$$

$$= 2 \int_1^5 \frac{\sqrt{t}}{2} dt = \int_1^5 \sqrt{t} dt = \frac{2}{3} t^{3/2} \Big|_1^5 =$$

$$= x^2 \sqrt{1+x^2} \Big|_0^2 - \frac{2}{3} t^{3/2} \Big|_1^5 =$$

$$= (2^2 \sqrt{1+2^2} - 0) - \frac{2}{3} (5^{3/2} - 1^{3/2}) =$$

$$= 4\sqrt{5} - \frac{2}{3} (5\sqrt{5} - 1) = 4\sqrt{5} - \frac{10\sqrt{5}}{3} + \frac{2}{3} =$$

$$= \frac{2\sqrt{5}}{3} + \frac{2}{3}$$

9.1.104

$$\int_0^{\pi/2} \frac{x \cos x}{\sin^2 x} dx = \left[\begin{array}{l} u = x \quad du = dx \\ dv = \frac{\cos x}{\sin^2 x} dx \quad v = \int \frac{\cos x}{\sin^2 x} dx = \end{array} \right]$$

$$= \left[\begin{array}{l} t = \sin x \\ dt = \cos x dx \end{array} \right] = \int \frac{t^{-1}}{t^2} dt = \frac{t^{-1}}{-1} + C = -\frac{1}{t} + C =$$

$$\begin{aligned}
 &= -\frac{1}{\sin x} + C \Big] = -x \frac{1}{\sin x} \Big|_{\pi/3}^{\pi/2} - \int_{\pi/3}^{\pi/2} -\frac{1}{\sin x} dx = \\
 &= \left[(1) \int_{\pi/3}^{\pi/2} \frac{1}{\sin x} dx = \int_{\pi/3}^{\pi/2} \frac{1}{\sin x} \cdot \frac{\sin x}{\sin x} dx = \int_{\pi/3}^{\pi/2} \frac{\sin x}{\sin^2 x} dx = \right. \\
 &= \int_{\pi/3}^{\pi/2} \frac{\sin x}{1 - \cos^2 x} dx = \left[\begin{array}{l} t = \cos x \\ dt = -\sin x dx \end{array} \right. \quad \left. dx = -\frac{1}{\sin x} dt \quad \begin{array}{l} x = \frac{\pi}{2} \quad t = 0 \\ x = \pi/3 \quad t = 1/2 \end{array} \right] = \\
 &= \int_0^{1/2} -\frac{dt}{1-t^2} = \int_0^{1/2} \frac{dt}{t^2-1} = \frac{1}{2} \ln \left| \frac{t-1}{t+1} \right| \Big|_0^{1/2} = \\
 &= -\frac{x}{\sin x} \Big|_{\pi/3}^{\pi/2} + \frac{1}{2} \ln \left| \frac{t-1}{t+1} \right| \Big|_0^{1/2} = \\
 &= -\left(\frac{\pi/2}{\sin \pi/2} - \frac{\pi/3}{\sin \pi/3} \right) + \frac{1}{2} \left(\ln \left| \frac{1/2-1}{1/2+1} \right| - \ln \left| \frac{0-1}{0+1} \right| \right) = \\
 &= -\left(\frac{\pi}{2} - \frac{2\pi\sqrt{3}}{9} \right) + \frac{1}{2} (-\ln 3) = \frac{2\pi\sqrt{3}}{9} - \frac{\pi}{2} - \frac{\ln 3}{2} = \\
 &= \frac{2\pi\sqrt{3}}{9} + \frac{\pi + \ln 3}{2}
 \end{aligned}$$

9.1.107

$$\begin{aligned}
 \int_0^{\pi/2} \sin \sqrt{x} dx &= \left[\begin{array}{l} t^2 = x \quad dx = 2t dt \\ x = \pi/4 \quad t = \pi/2 \end{array} \right. \quad \left. x=0 \quad t=0 \right] = \\
 &= \int_0^{\pi/2} 2t \sin t dt = \left[\begin{array}{l} u = t \quad du = dt \\ dv = \sin t dt \quad v = -\cos t \end{array} \right] = \\
 &= 2 \left(-t \cos t \Big|_0^{\pi/2} - \int_0^{\pi/2} -\cos t dt \right) = 2 \left(-t \cos t \Big|_0^{\pi/2} + \sin t \Big|_0^{\pi/2} \right) = \\
 &= 2 \left(-\left(\frac{\pi}{2} \cos \frac{\pi}{2} - 0 \right) + (\sin \frac{\pi}{2} - \sin 0) \right) = 2(0 + 1 - 0) = 1 \cdot 2 = 2
 \end{aligned}$$

9.1.108

$$\begin{aligned}
 \int_0^9 e^{\sqrt{x}} dx &= \left[\begin{array}{l} t^2 = x \quad dx = 2t dt \\ x=9 \quad x=0 \\ t=3 \quad t=0 \end{array} \right] = \int_0^3 e^t \cdot 2t dt = \\
 &= 2 \int_0^3 t \cdot e^t dt = \left[\begin{array}{l} u=t \quad du=dt \\ dv=e^t dt \quad v=e^t \end{array} \right] = 2 \left(t \cdot e^t \Big|_0^3 - \int_0^3 e^t dt \right) = \\
 &= 2 \left(t \cdot e^t \Big|_0^3 - e^t \Big|_0^3 \right) = 2 \left((3 \cdot e^3 - 0) - (e^3 - e^0) \right) = \\
 &= 2 \left(3e^3 - e^3 + 1 \right) = 2(2e^3 + 1)
 \end{aligned}$$